



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-3)

– Hira Kumar

Control Structures

- In programming, control structures/ Instructions decide the *flow of execution*—that is, which statements run, when they run, and how many times they run.

There are Four types of control structures:

1. Sequence Statements
2. Selection Statements (Decision making/Conditional statements)
3. Looping (Iteration)
4. Jump Statements

1. Sequence Control Structure

- This is the default flow of a program.
- Statements execute one after another, from top to bottom, in the order they are written.
- Each step runs sequentially—no skipping, no repeating.

Example :

```
#include<iostream>
#include<conio.h>
using namespace std;
int main()
{
    cout << " Hello Hira!" << endl;
    cout << " Let's learn C++ " << endl;
    cout << " Thank You !";
    getch();
}
```

C:\Users\hira\Desktop\hello.exe

```
Hello Hira!
Let's learn C++
Thank You !
```

2. Selection (Decision-Making) Control Structure

- Here, the program chooses between different paths based on a condition.
- If a condition is true → one block runs & If false → another block runs.
- e.g. →
 - if
 - if-else
 - if-else ladder
 - switch

3. Iteration (Looping) Control Structure

- This is used when a block of code must be repeated multiple times.
- The same statements run again and again until a condition becomes false.
- e.g. →
 - for
 - while
 - do-while

4. Jump Statements

- A jump statement is a statement that immediately changes the normal flow of program execution by transferring control from one place to another.
- e.g →
 - break
 - continue
 - goto
 - return

In Details

Selection / Decision making Statements

- The conditional statements (also known as decision making control structures) such as if, if else etc. are used for decision-making purposes in C++ programs.
- They are also known as Decision-Making Statements and are used to evaluate one or more conditions and make the decision whether to execute a set of statements or not.
- These decision-making statements in programming languages decide the direction of the flow of program execution.

❖ **Need of Conditional Statements**

- ✓ There come situations in real life when we need to make some decisions and based on these decisions, we decide what should we do next.
- ✓ Similar situations arise in programming also where we need to make some decisions and based on these decisions we will execute the next block of code.

Types of Conditional Statements in C++

- I. if statement
- II. if-else statement
- III. nested if-else statements
- IV. else-if ladder

1. if

- The if statement is the most simple decision-making statement.
- It is used to decide whether a certain statement or block of statements will be executed or not i.e if a certain condition is true then a block of statements is executed otherwise not.

Syntax :-

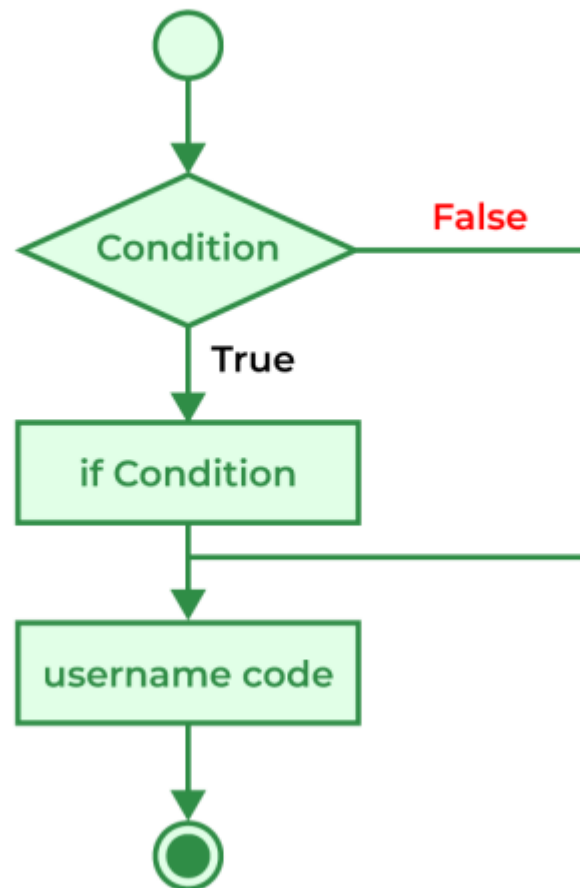
if(condition)

{

Statements execute ; // if condition is true

}

Flowchart:



Question

Q1. Write a program in c++,in which Check person can apply for Driving licence or not.

Method 1.

```
#include<iostream>
#include<conio.h>
using namespace std;
int main()
{
    int age;
    cout << "Enter Your age : ";
    cin >> age;
    if (age>=18)
    {
        cout << " you are eligible for DL" << endl;
    }
    else
    {
        cout << " Your are not eligible for DL" << endl;
    }
    getch();
}
```

```
C:\Users\hira\Desktop\hello.exe
Enter Your age : 20
you are eligible for DL
```

Method 2.

```
#include<iostream>
#include<conio.h>
using namespace std;
int main()
{
    int age;
    cout << "Enter Your age : ";
    cin >> age;
    if (age>=18)
        cout << " you are eligible for DL" << endl;
    if (age<18)
        cout << " Your are not eligible for DL" << endl;
    getch();
}
```

C:\Users\hira\Desktop\hello.exe
Enter Your age : 17
Your are not eligible for DL

Method 3.

```
#include<iostream>
#include<conio.h>
using namespace std;
int main()
{
    int age;
    cout << "Enter Your age : ";
    cin >> age;
    cout << ((age>=18)? "you are eligible" : "you are not eligible");
    getch();
}
```

C:\Users\hira\Desktop\hello.exe
Enter Your age : 20
you are eligible

- Q2.** Write a program in c++,in which Check person can eligible for vote or not. (Using only if statement)
- Q3.** Write a program in c++ take any number and check this number is Odd or Even. (Using only if statement)
- Q4.** Write a program in c++,in which Check Student pass or fail.(Using only if statement)
- Q5.** Write a program in c++ to verify whether an ATM withdrawal can be done based on available balance using only if statement.
- Q6.**Write a program to check the temperature condition of the day (hot or normal) using only if statement.
- Q7.** Write a program take two number and check greatest,smallest or equal.
- Q8.** Write a c++ program that takes the session year and the current year as input and tells which year of study the student is in.

2. if-else in C++

- The if statement alone tells us that if a condition is true it will execute a block of statements and if the condition is false it won't.

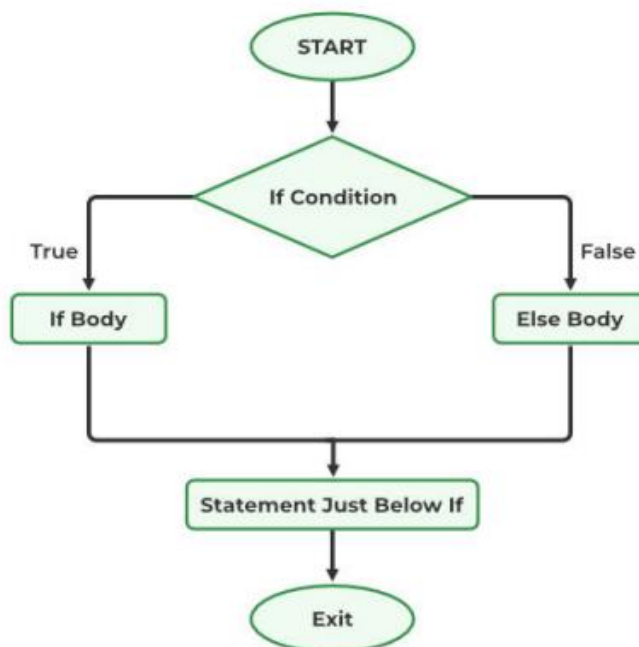
But what if we want to do something else when the condition is false?

- Here comes the C++ else statement. We can use the else statement with the if statement to execute a block of code when the condition is false.
- The if-else statement consists of two blocks, one for false expression and one for true expression.

Syntax :

```
if (condition)
{
    // Executes this block; // if condition is true
}
else
{
    // Executes this block; // if condition is false
}
```

Flowchart:



Question

Q9. Write a program in c++, in which compare two numbers and print which one is greater.

Q10. Write a program in c++, in which check whether a given number is even or odd using bitwise operators.

Q11. Write a program in c++, in which Create a login validation program.

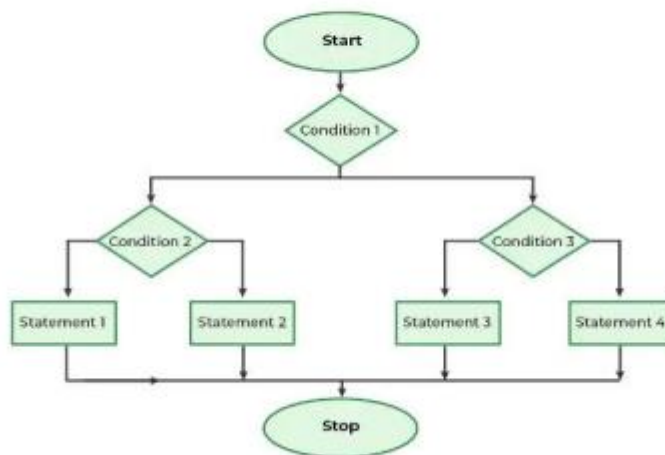
3. Nested if-else in C++

- A nested if in C++ allow us to nested if statements within if statements, i.e, we can place an if statement inside another if statement.

Syntax :

```
if (condition1)
{
    // Executes when condition1 is true
}
if (condition2)
{
    // Executes when condition2 is true
}
else
{
    // Executes when condition2 is false
}
```

Flowchart:



Question

Q12. Write a c++ program to input marks of a student and display the grade using nested if-else.

Q13. Write a c++ program to input three numbers and find the largest number using nested if-else.

Q14. Write a c++ program to check whether a year is a leap year or not using nested if-else.

Q15. Write a c++ program to input age and determine the ticket category (child,adult,senior) using nested if-else.

Q16. Write a c++ program to input a character and check whether it is a vowel, consonant,digit or special symbol using nexted if-else.

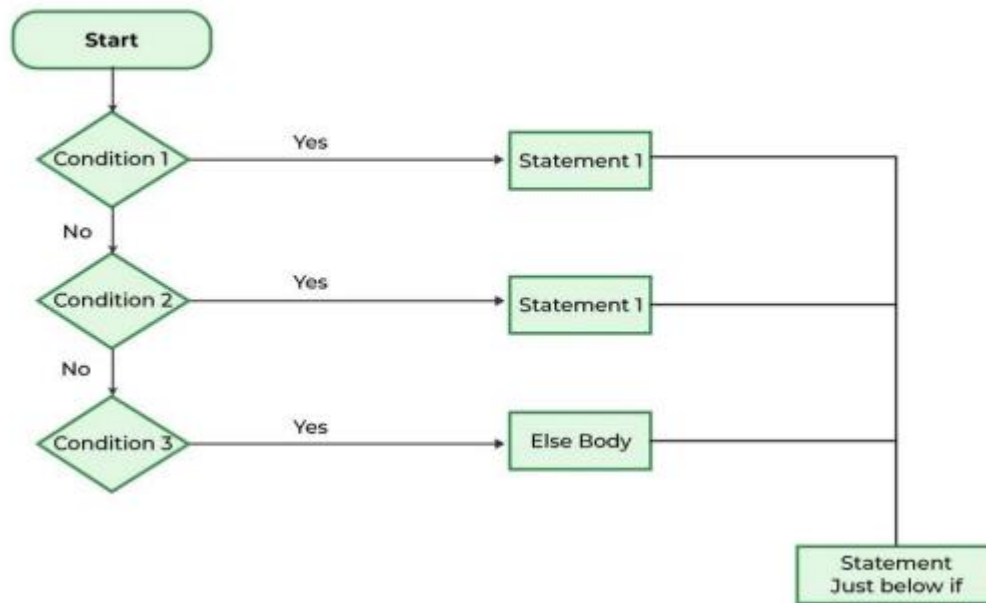
4. if-else-if Ladder in C++

- The if else if statements are used when the user has to decide among multiple options.
- The C++ if statements are executed from the top down. As soon as one of the conditions controlling the if is true, the statement associated with that if is executed, and the rest of the C++ else-if ladder is bypassed.
- If none of the conditions is true, then the final else statement will be executed. if-else-if ladder is similar to the switch statement.

Syntax :

```
if (condition)
    statement;
else if (condition)
    statement;
else if (condition)
    statement;
else if (condition)
    statement;
..
..
..
..else
    statement;
```


Flowchart:



Question

Q16. Write a c++ program to read marks of a student and print the 1st/2nd/3rd or grade using if-else-if ladder.

Q17. Write a c++ program to find the largest among three numbers using if-else-if ladder.

Q18. Write a c++ program to categorize electricity usage as low, medium, or high if-else-if ladder.

Q19. Write a c++ program to determine whether a person is a minor, adult, or senior citizen using if-else-if ladder.

Q20. Write a c++ program to check whether a number is positive, negative or zero using if-else-if ladder.

=====HIRA KUMAR=====